

Smart Parking in Smart City

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Abstract - Parking resources efficient management in modern urban environment is getting more important because parking related activities are resource consuming in terms of spent time, fuel, carbon emission, air pollution and motorists frustration. According to research studies [15] average 30% of cars in urban areas are running around in search for available parking lot. This paper highlights challenges and issues related to the optimal parking resources management in smart city environment trying to give comprehensive overview including all elements relevant for optimal parking resources management. Mathematical, theoretical models and game theory concepts will be considered targeting parking specific challenges and issues. Democratization in parking resources management and commercialization will be considered and proposed focused to social welfare. Digital Parking platform will be proposed and described offering game change and democratized parking resource management. New models for further parking research will be proposed.

I. INTRODUCTION

Parking lot allocation in modern cities is always real challenge for every motorist. Usually there are business district zones where many potential clients want to park their vehicles. Many papers analyzing this are proposing as solution digital, smart parking platform introducing negotiation process between user Agent (UA) held by motorist and Parking Manager (PA) managing all parking resources in urban, smart city environment [14]. This approach is step forward from traditional parking organization towards parking resources optimal management but we have to consider few more details in order to have comprehensive perception of challenges related to optimal and efficient parking resources management in smart city environment.

Reaching parking set having available parking lots by motorist at zero cost or non-zero cost can be analyzed in view of non-cooperative multi-player game. In order to achieve any type of Nash equilibrium [3] among motorist playing game of parking every single of them has to reach at least one parking set containing available parking lot at any cost.

Once when motorist reached any parking set every player has same strategy. To circle around parking place in one of available directions looking for available parking lot. In that situation motorist goes into classic „Linear search problem“ game during [1] looking for specific parking lot inside parking set.

In traditional population behavior model we have car owner parked on parking place in private garage or private parking lot in front of building. During usual business day car owner drives to another part of the city and looking for parking place too. Described behavior during working day generates high percentage of private

owned parking lots in garages or on streets close to residential buildings dis-allocated and turned off from parking game.

In every urban area there is one more major player in game of parking. Very often public owned company „Parking Service Company“ (PSCo). PSCo major sources of income are:

a) managed parking lots renting on 1 hour time frame basis;

b) penalty tickets issued to inappropriate parking vehicles including but not limited to vehicles locking, vehicles moving to dedicated parking space in order to charge not just penalty ticket but transportation service too. PSCo company having such income structure is not strongly motivated to work permanently on parking resources management and improvement because major income comes from Nash equilibrium which is not Pareto optimal [3].

II. GAME OF PARKING

In order to make contribution towards parking game optimization all relevant elements has to be identified in Smart City environment with micro-navigation enabled infrastructure.

Sum of parking lots owned by urban area government/authorities/ PSCo with non-zero cost available can be marked as

$$\text{SUM}(\text{UA}_i) \{i=1,n\}$$

UA_i is single parking set with finite parking lot number.

Set of parking places available in the private domain/private owners can be marked as

$$\text{SUM}(\text{PO}_j) \{j=1,m\}$$

PO_j is single parking set with finite parking lot number.

Zero cost available parking lots in smart city zone can be marked by

$$\text{SUM}(\text{ZC}_k) \{k=1,q\}.$$

Total number of parking lots available in the Smart City area is:

$$\text{TPL} = \text{SUM}(\text{UA}_i) + \text{SUM}(\text{PO}_j) + \text{SUM}(\text{ZC}_k)$$

Taking into account that in traditional private owner behavior parking lots expressed by $\text{SUM}(\text{PO}_j)$ are not visible by end users so we can consider that existing parking game has

$$\text{SUM}(\text{PO}_j) = 0.$$

Zero cost parking lots are usually not marked, not regulated, not promoted because this is not in best PSCo

financial interest and private companies and individuals are not allowed to manage them.

In traditional behavior motorists searching for available parking lost can count on

$$\text{SUM}(ZCk \cdot vk) \{ 0 < vk < 1 \}$$

parking lots reachable in game of parking.

Number of cars requiring parking during day can be marked as TC. Numbers of cars requesting parking place is total sum of cars owned by residents RC increased for total number of urban zone visitors VC deducted for total number of residential cars left analyzed urban zone LC. So total number of cars looking for parking lot available in smart city area is:

$$TC = RC + VC - LC$$

with one or more lots available.

To complete parking game elements we have to consider vehicles „in move“. Vehicles running from one location to another occupying Smart City streets. We can mark those cars by SC.

$$TC = RC + VC - LC - SC$$

III. PARKING DEMOCRATIZATION

According to Donald Shoup's PhD research [15] 08.00% to 75.00% of total city traffic is generated by vehicles tracking down free parking lot. According to same research average 30.00% vehicles are running around searching for free parking lot.

Time consumed by drivers searching for free parking lot is expense for drivers personally, employers and society in general. CO2 emission is increased and total national resources are decreased because of non-optimal parking lot allocation.



Figure 1. xZone smart city platform adds new parking lot into game

According to mentioned in introduction of this paper PSCo are not strongly motivated to increase quality of parking service nor to optimize it. Pareto-non-efficient Nash equilibrium in parking game generates better

financial results for PSCo then Pareto-efficient Nash equilibrium.

In order to increase parking quality level in Smart City environment we developed Smart City platform supporting advanced smart parking features:

a) Rent-a-Parking giving opportunity to small parking set owners {1..20} to make own parking set available on market (Figure 1). xZone platform is playing role of Parking Manager [14] and gives opportunity to xZone smart phone application playing role of User Agent Manager (UAM) [14] to publish new available parking set on market. While owner is willing to be part of the xZone Smart City platform all income generated from parking set renting is transferring to parking set owner account. Parking Set owner can be individual, company, or building council.

At the end of day UAM generates daily report regarding market effect of the parking set (Figure 2).

Short simplified algorithm describing Rent-a-Parking feature:

1>>>John Smith can buy xCube smart city building block device in regular shop or order on the Internet.

2>>>John Smith can install xCube device in garage or owned parking lot/lots together with parking lot car presence sensor/sensors.

3>>>John Smith can download android smart phone application from "Play Store". Application activation process will initialize dedicated account number for submitted xCube ID. Geographic elements related to parking lots and number of lots will be associated to xCubeID and registered on the xCube Smart City platform.

4>>When is applicable, John will can start application and make owned parking lots "visible" on the market.

5>>xCube Smart City platform is running sales process as Parking Manager and keep John Smith informed via smart phone application push messages.

6>>Depends on need John Smith can make parking lots "invisible" to the market again with setting adjustments on smart phone application.

7>>John Smith can decide how earned money will be used. For mPayments via xZone smart phone application or he can just with draw money to specific Visa/Mastercard credit card

b) Sophisticated low-cost parking sensor available to monitor up to 10 parking lots in street-line or garage without need for any additional infrastructure work like drilling holes into ground and additional wiring.

By simple installation all marked parking lots will be controllable by Parking Manager (PM) and Smart City users will be aware via xZone platform that place is ready for reservation.

Existing parking sensors are slowing down smart parking deployment in Smart City environment because deployment cost per single parking lot goes between (20-50 USD).

With 10 times lower cost xCube new generation parking sensor will speed-up smart parking features deployment in urban areas.

PSCo and authorities will not be very interested in increasing market in described way since proposed approach is going to decrease sum of financial income coming from issued parking tickets.

Parking democratization has to go towards total decentralization of the Smart City parking lot management. According to proposed rules every interested party will be able to organize and maintain existing zero-cost parking lots what will change vk value into 1 in expression:

$$\text{SUM}(ZCk*vk) \{ 0 < vk < 1 \}$$

for every single parking set belonging to zero-cost parking set. This action will transfer ZC parking set into PO parking set type what will be additional step towards total Smart Parking in Smart City environment.

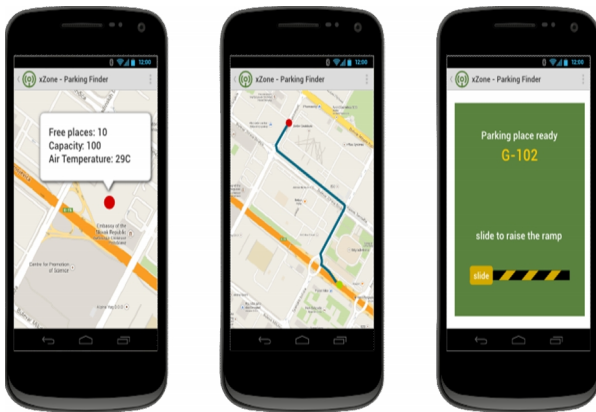


Figure 2. xZone application in User Agent Manager role



Figure 3. xZone Smart City platform daily report

Interested parties will be strongly motivated to increase sum of PO parking sets. City authorities will have additional income in shared profit cooperation model.

All mentioned will move game towards easy reachable Strong-Nash equilibrium in Smart City game of

parking. Strong Nash equilibrium has to be Pareto-efficient.

Thanks to totally smart environment minimal time frame for parking lot rent can go down to ½ or ¼ hours what will increase parking system dynamics.

Smart City with described Smart Parking system development approach will push additionally community towards social welfare community.

IV. CONCLUSION

New Smart City features proposed in this paper related to further smart city urban development, not only technological but social too can help Smart City environment to become not just technologically progressive but advanced in terms of society welfare. Innovative business processes modeling based on Cross Loyalty, Smart Mobility and Smart City features can push smart city community towards 21st century community (efficient, motivated and resources limitation aware).

New generation of smart parking sensors, elegant, easy to deploy and low-cost will increase transformation zero-cost type of parking sets into private owned/controlled non-zero cost type of parking space:

$$ZC \rightarrow PO$$

what will increase Smart City efficiency and motivation.

In existing studies and papers is hard to identify models taking into account number of vehicles on the streets running to another part of the city or number of vehicles approached to parking set area but not used any parking lot.

Proposal to other researches is to make research based on modified Nagel-Schreckenberg model in order to estimate number of vehicles approaching to the parking set. Data for Nagel-Schreckenberg modification should be gathered using sophisticated movement detection sensors in order to include real data into model [18].

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